

EDA Report

Executive Summary

The Social Engine Posts dataset contains 12,000 social media posts from 1,500 users, covering the period from 1 May 2024 to 30 April 2025. The dataset includes post identifiers, user identifiers, platform information, text content, timestamps, and three engagement metrics: likes, shares, and comments.

Exploratory Data Analysis was performed on the cleaned dataset to understand its data composition, engagement behaviour, platform-level patterns, temporal activity, user-level posting behaviour, relationships between engagement metrics, and statistical anomalies.

The analysis shows that the five identified platforms have a relatively balanced representation, while 14.87% of posts have missing platform information. Engagement values are broadly distributed across their valid ranges, with likes ranging from 0–5,000, shares from 0–2,000, and comments from 0–1,000. The three engagement metrics show almost no linear correlation, indicating that high performance in one engagement metric does not necessarily correspond to high performance in another.

Posting activity remains relatively stable across the year, although February 2025 records the lowest monthly post volume. Average engagement also remains relatively stable over time. User activity is broadly distributed across all 1,500 users, with an average and median of 8 posts per user.

The analysis also identified a substantial concentration of timestamps at 00:00, requiring caution when interpreting hourly posting behaviour because some source timestamps originally contained date information without an explicit time. Finally, no IQR-based statistical outliers were identified in the engagement metrics after data cleaning.

Overall, the EDA establishes the structure and behavioural characteristics of the cleaned dataset while highlighting important data-quality considerations that should be accounted for in subsequent analysis.

Dataset Overview and Data types

Dataset Overview

- Total posts: 12,000
- Total users: 1,500
- Columns: 8
- Observation period: 1 May 2024 – 30 April 2025
- Platforms: Facebook, YouTube, Twitter, Reddit, Instagram
- Engagement metrics: Likes, Shares, Comments
- Missing platform values: 1,784 (14.87%)
- Missing text values: 1,688 (14.07%)
- Missing likes: 2,323 (19.36%)
- Duplicate post IDs: 0
- Exact duplicate rows: 0
- Invalid user IDs: 0

Data Types

- post_id, user_id, platform, text_content → categorical/text
- timestamp → datetime
- likes, shares, comments → numerical

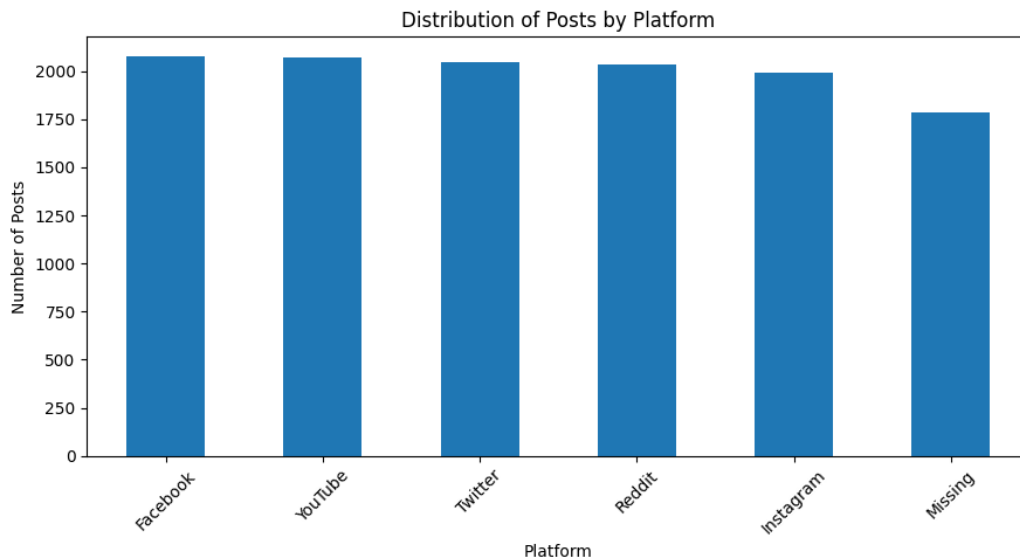
Data Quality & Cleaning Summary

- Duplicate rows: 0 exact duplicates
- Duplicate post IDs: 0
- Missing platform: 1,784 (14.87%)
- Missing text content: 1,688 (14.07%)
- Missing likes: 2,323 (19.36%)
- Negative likes detected: 509
- Negative likes corrected: 509
- Negative shares/comments: 0
- Invalid user IDs: 0
- Missing timestamps: 0
- Timestamp range after standardisation: 1 May 2024 – 30 April 2025
- HTML-like tags/entities: identified and cleaned
- Unrecoverable missing values: retained as NaN rather than fabricated

Cleaning principle: Only values that could be reliably corrected were modified; unrecoverable information was preserved as missing.

Platform Distribution

- Facebook: 2,074 posts (17.28%)
- YouTube: 2,073 posts (17.27%)
- Twitter: 2,049 posts (17.08%)
- Reddit: 2,031 posts (16.92%)
- Instagram: 1,989 posts (16.58%)
- Missing platform: 1,784 posts (14.87%)



- The five known platforms have a highly balanced representation.
- No single platform dominates the dataset.
- Missing platform records are substantial and should be treated separately in platform-level analysis.

Engagement Distribution

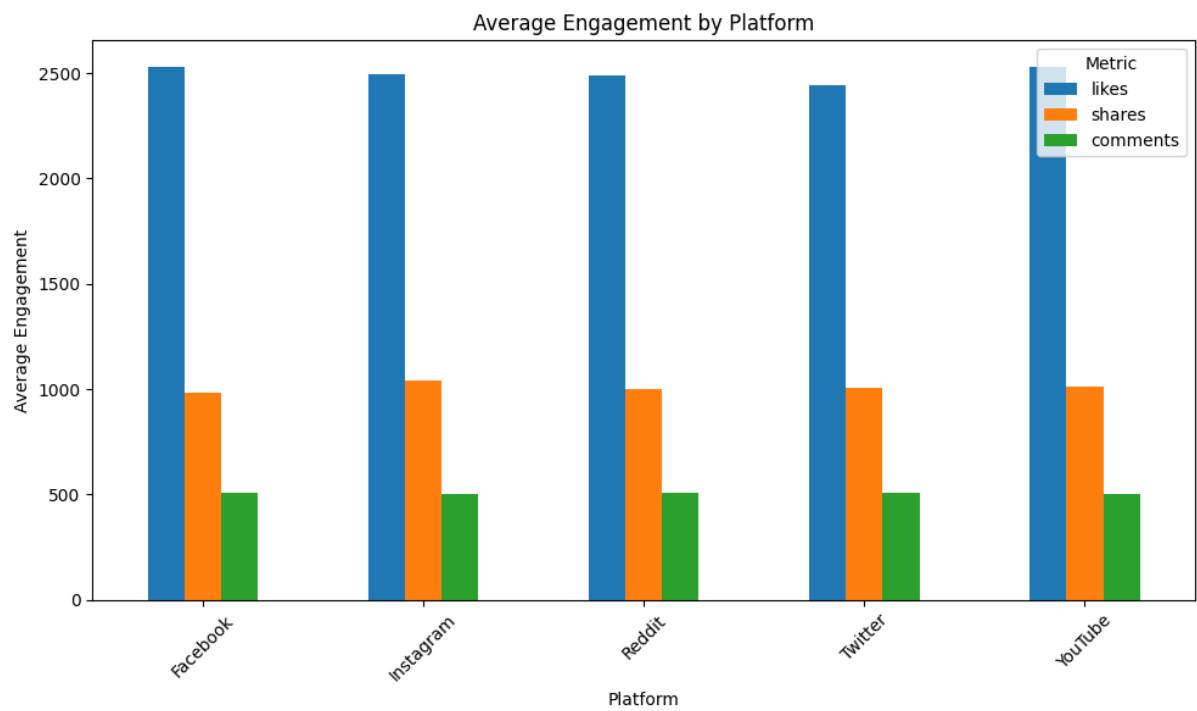
	count	mean	std	min	25%	50%	75%	max
likes	9677.0	2493.505115	1438.286521	0.0	1240.0	2502.0	3724.0	5000.0
shares	12000.0	1007.167167	575.072282	0.0	510.0	1018.0	1501.0	2000.0
comments	12000.0	504.345750	288.684160	0.0	253.0	503.0	755.0	1000.0

- Engagement values are broadly distributed across their respective ranges.
- No major concentration or unusual spike is visible in the distributions.
- 2,323 likes values are missing and were excluded from likes-based distribution analysis.
- Missing likes were not treated as zero, preserving the distinction between unavailable and actual zero engagement.

Engagement by Platform

	likes		shares		comments	
	mean	median	mean	median	mean	median
platform						
Facebook	2526.55	2573.0	984.17	993.5	506.94	504.0
Instagram	2494.51	2495.0	1040.84	1049.0	499.80	498.0
Reddit	2487.80	2496.0	1002.23	1016.0	511.18	513.0
Twitter	2439.87	2443.5	1005.39	1022.0	506.13	504.0
YouTube	2528.33	2619.5	1011.83	1017.0	504.38	498.0

- Highest average likes: YouTube — 2,528.33
- Highest average shares: Instagram — 1,040.84
- Highest average comments: Reddit — 511.18
- Lowest average likes: Twitter — 2,439.87

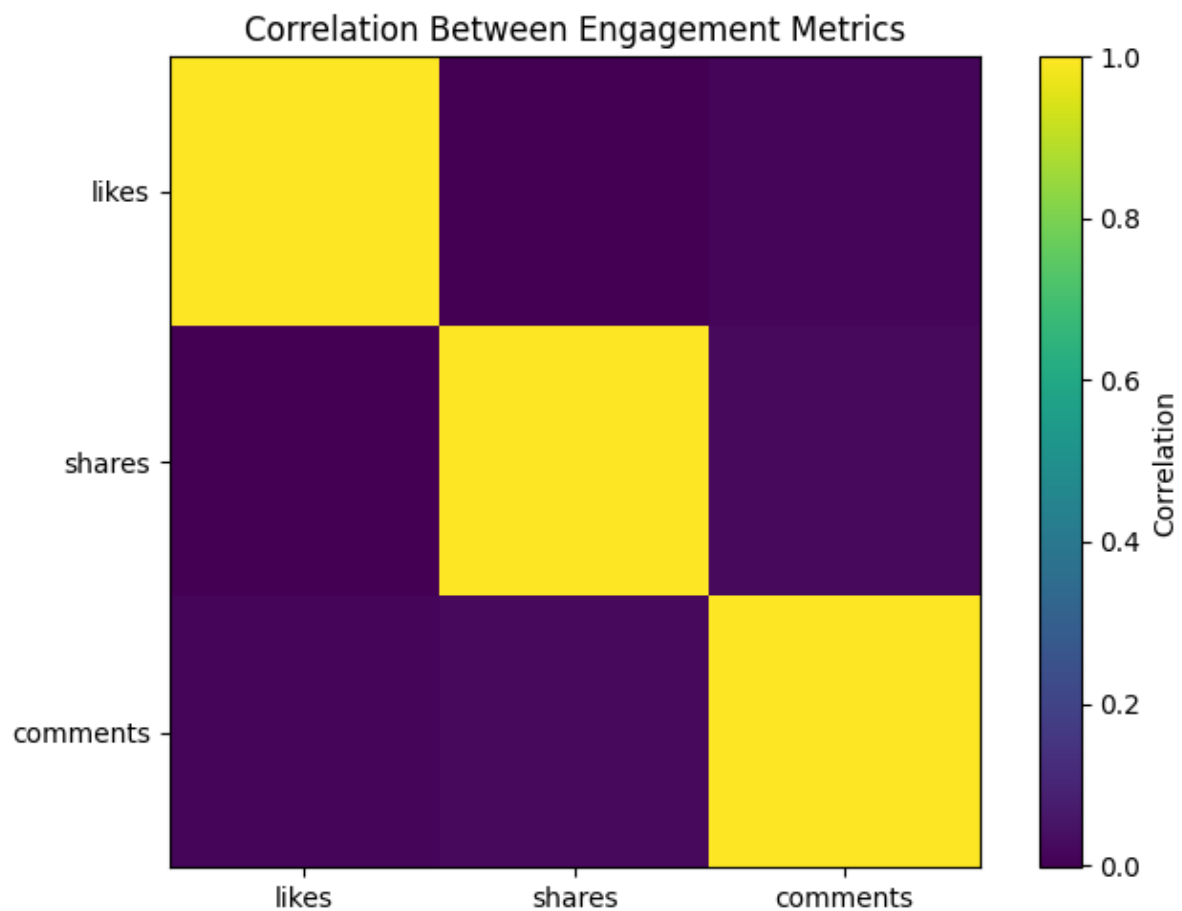


- No single platform consistently leads across all three engagement metrics.
- Platform differences are metric-specific, rather than indicating one universally most engaging platform.
- Missing-platform records were excluded from this comparison.

Correlation Between Engagement Metrics

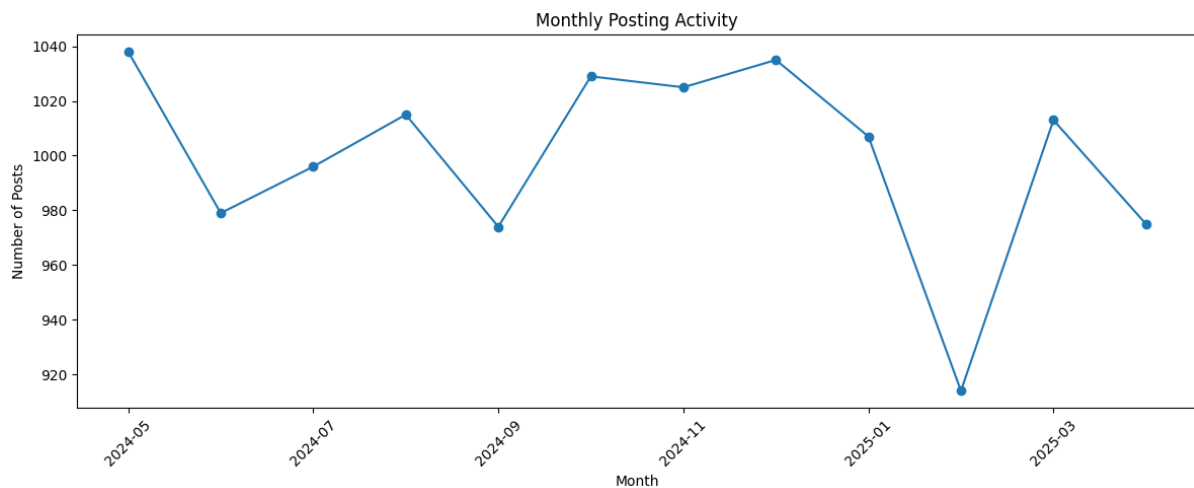
	likes	shares	comments
likes	1.000	-0.003	0.013
shares	-0.003	1.000	0.024
comments	0.013	0.024	1.000

- Likes vs Shares: Pearson correlation = -0.003
- Likes vs Comments: Pearson correlation = 0.013
- Shares vs Comments: Pearson correlation = 0.024



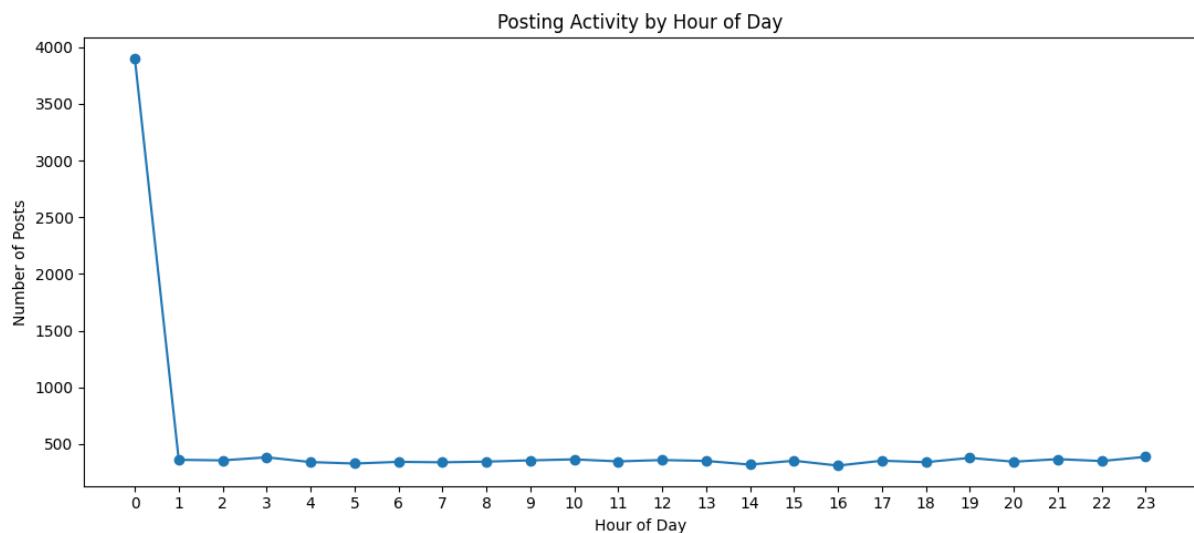
- All correlations are very close to zero.
- The metrics show almost no linear association with one another.
- High likes do not necessarily correspond to high shares or comments.
- Correlation indicates association, not causation.

Posting Activity Over Time



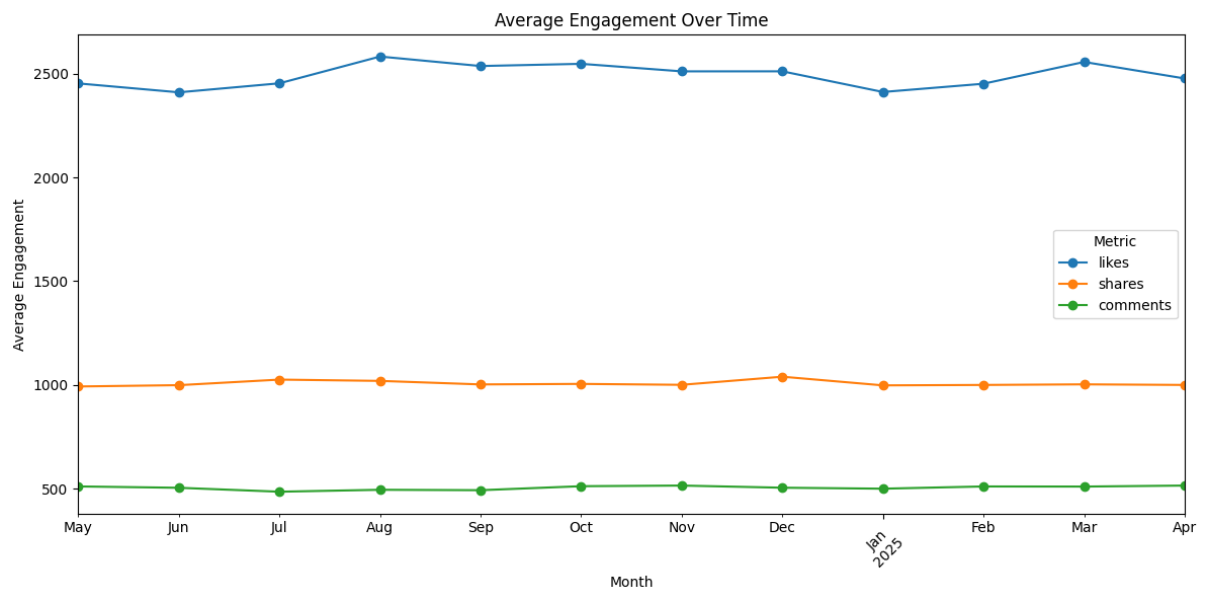
- The dataset covers 12 months from May 2024 to April 2025.
- Monthly post volume remains relatively stable, mostly between 975 and 1,038 posts.
- Highest: May 2024 — 1,038 posts
- Lowest: February 2025 — 914 posts
- Posting volume rises again to 1,013 posts in March 2025.
- No sustained upward or downward trend is observed.
- Overall activity shows moderate monthly fluctuations around a stable level.

Posting Activity by Hour



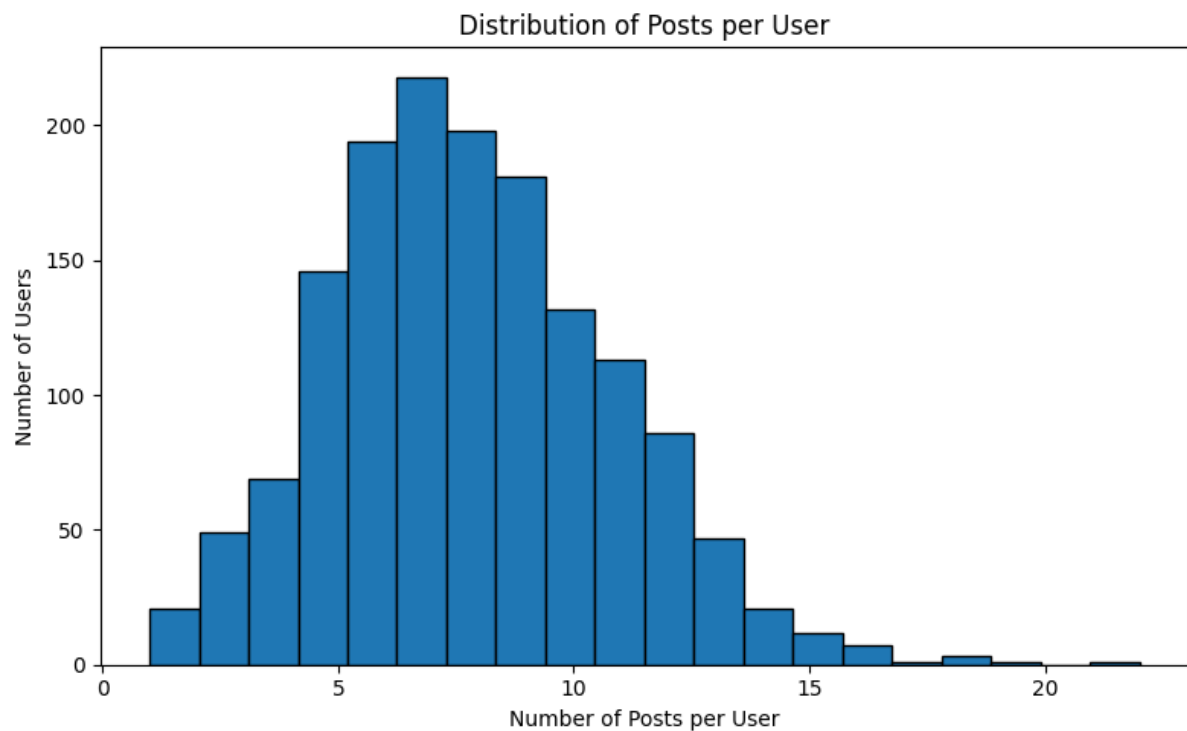
- 00:00: 3,902 posts — highest observed count.
- 16:00: 311 posts — lowest observed count.
- Most non-midnight hours show approximately 300–400 posts.
- The unusually high 00:00 count is likely influenced by date-only timestamps standardised to midnight.
- Therefore, 00:00 should not be interpreted as a genuine midnight posting preference.
- Hour-level behavioural analysis should be treated with caution due to incomplete time-of-day information.

Engagement Over Time



- Average likes remained between 2,410.80 and 2,583.40 across the 12 months.
- Average shares remained between 992.53 and 1,039.64.
- Average comments remained between 485.13 and 514.73.
- Monthly changes are relatively small, with no sustained increase or decrease.
- The largest monthly increase in average likes was 5.3% in August 2024.
- Overall, engagement levels remain relatively stable throughout the observation period.

User Posting Activity



- Total users: 1,500
- Average posts per user: 8
- Median posts per user: 8
- Posts per user: 1–22
- Middle 50% of users: 6–10 posts
- Most active user: 22 posts
- User activity is broadly distributed rather than heavily concentrated among a small group.
- All 1,500 users have at least one corresponding post.

Engagement Outlier Analysis

	Metric	Q1	Q3	IQR	Lower Bound	Upper Bound	Outlier Count	Outlier Percentage
0	likes	1240.0	3724.0	2484.0	-2486.0	7450.0	0	0.0
1	shares	510.0	1501.0	991.0	-976.5	2987.5	0	0.0
2	comments	253.0	755.0	502.0	-500.0	1508.0	0	0.0

- Likes: 0 IQR-based outliers
- Shares: 0 IQR-based outliers
- Comments: 0 IQR-based outliers
- All valid engagement values fall within their respective IQR-based bounds.
- No additional outlier removal was required.
- Extreme values were not automatically removed, since statistical extremeness alone does not indicate invalid data.
- The analysis confirms that engagement values are statistically contained within expected ranges after cleaning.

Key EDA Insights & Conclusion

Key Insights

- The dataset contains 12,000 posts from 1,500 users with complete user-level coverage.
- The five known platforms have relatively balanced post volumes, with 14.87% platform values missing.
- Engagement metrics show almost no linear correlation with one another.
- No single platform consistently performs highest across likes, shares, and comments.
- Monthly posting volume and average engagement remain relatively stable over the 12-month period.
- User activity is broadly distributed, with a mean and median of 8 posts per user.
- Hourly analysis is affected by the large number of timestamps recorded at 00:00, so time-of-day conclusions require caution.
- No IQR-based outliers were identified in the engagement metrics.

Conclusion

The EDA provides a clear understanding of the dataset's structure, engagement behaviour, platform distribution, temporal patterns, and user activity. The cleaning process resolved identifiable data-quality issues while preserving information that could not be reliably recovered. Overall, the cleaned dataset is suitable for further analytical and data-engineering workflows, with missing values and timestamp limitations kept in consideration during downstream analysis.

